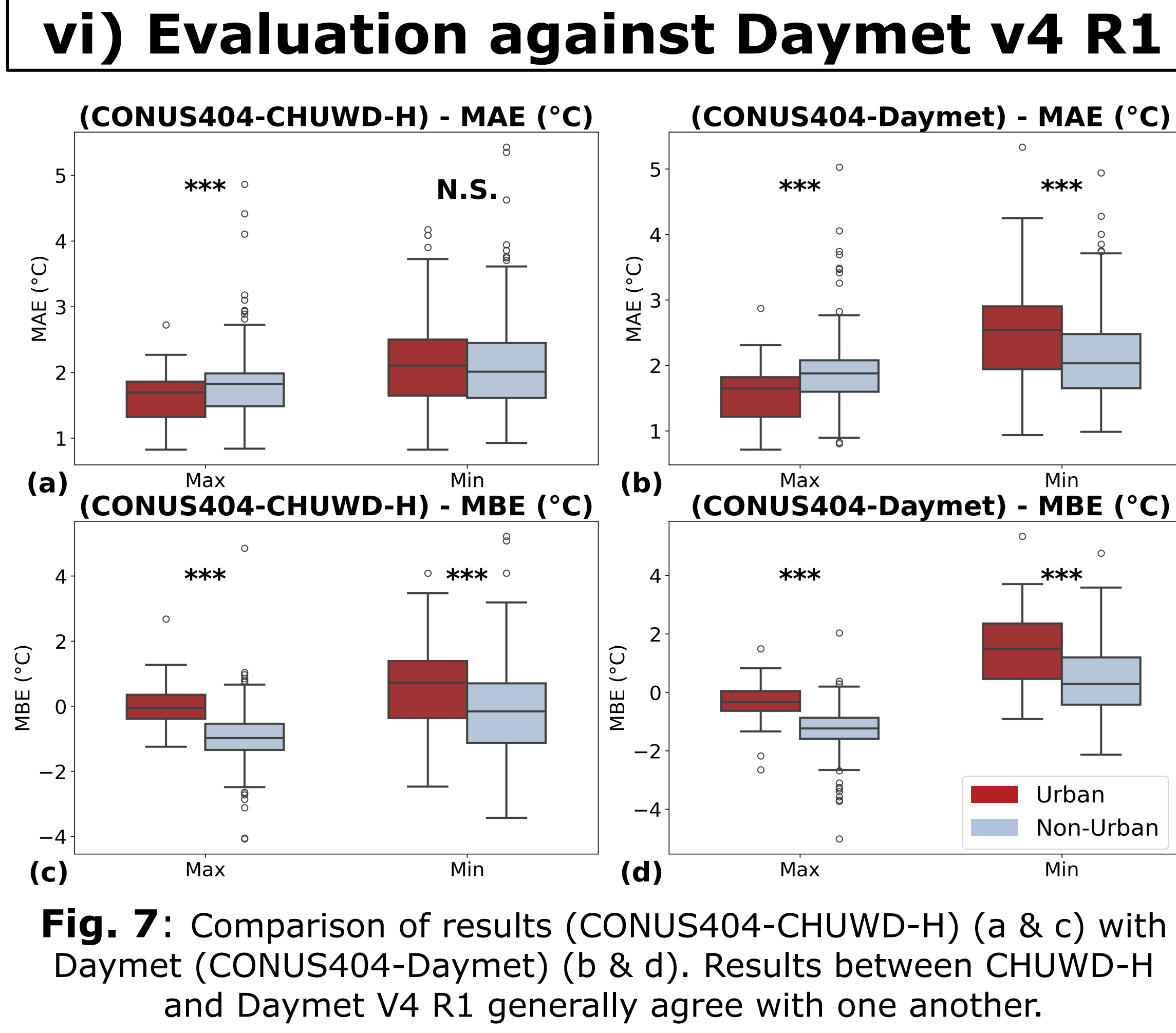
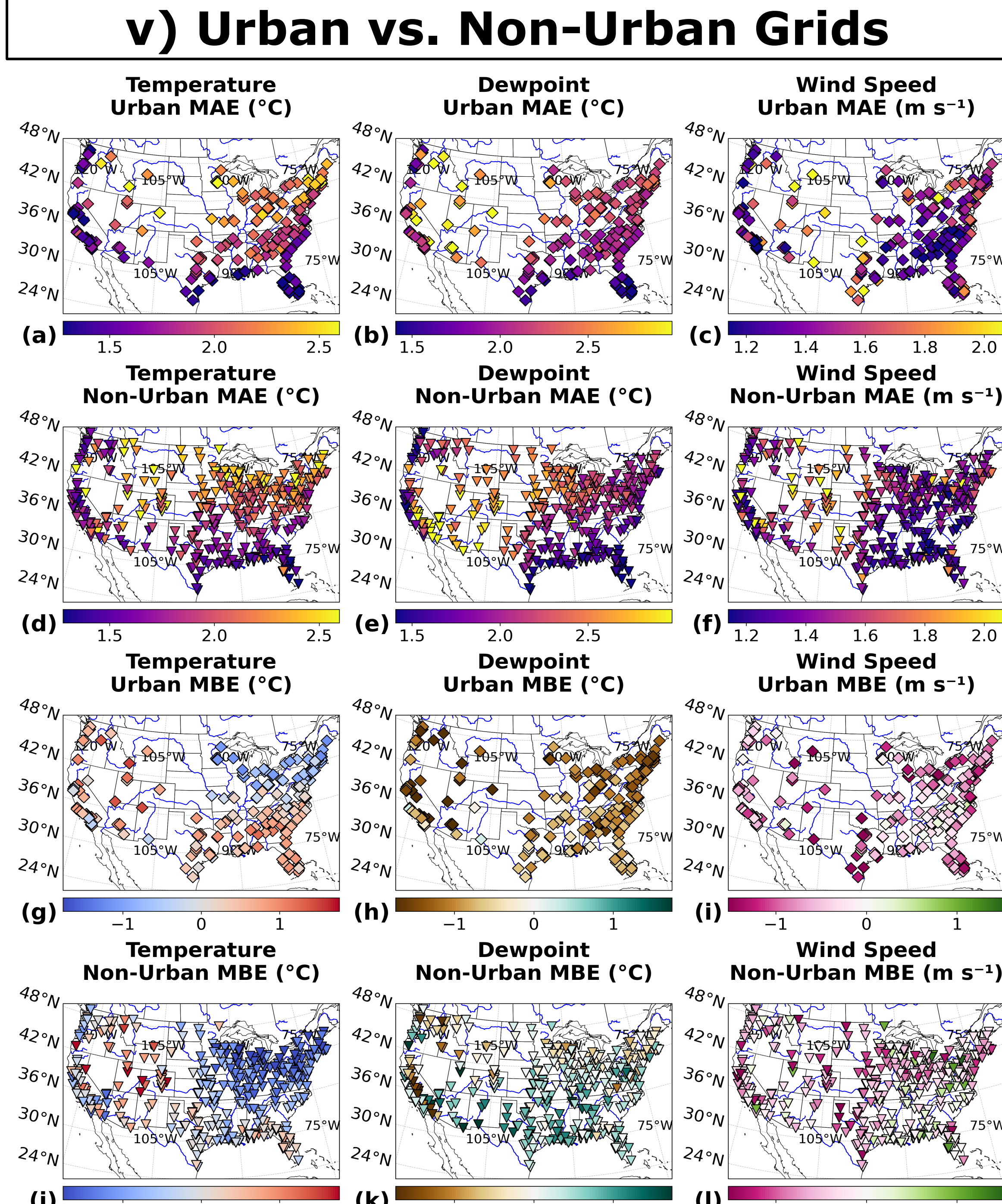
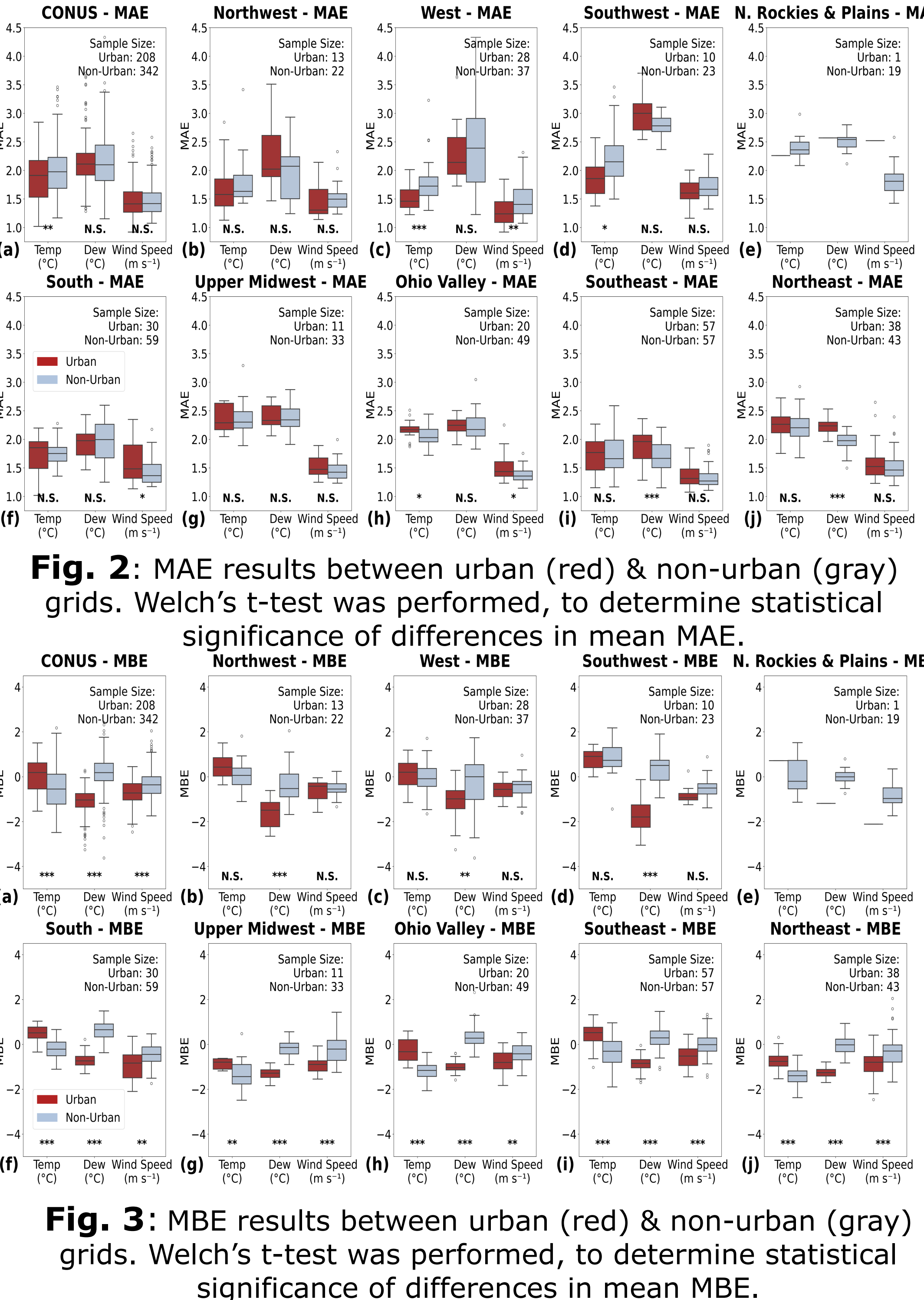
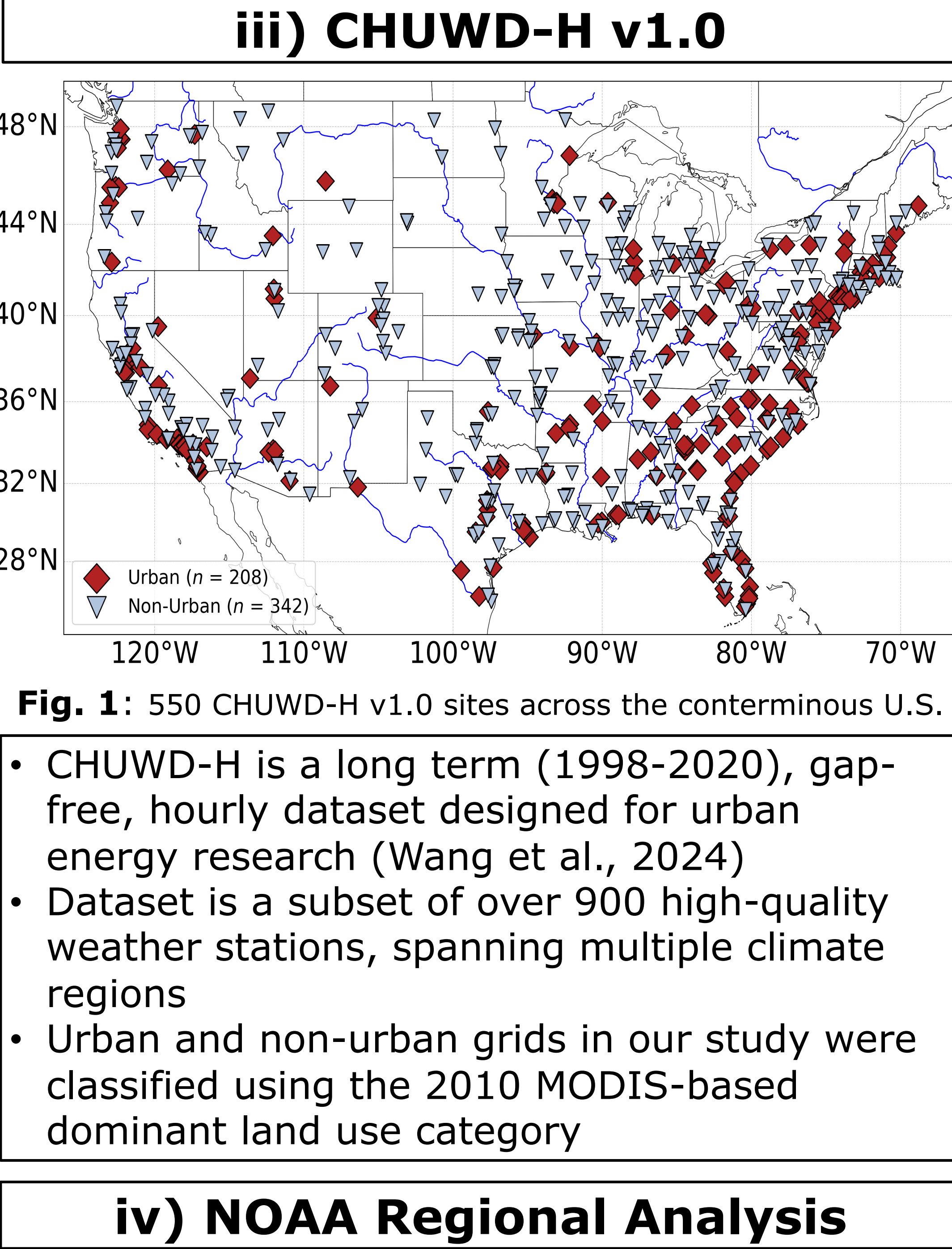


i) Background

- CONUS404 is one of the highest resolution (4-km) convection-permitting reanalysis datasets (Rasmussen et al., 2023), which could allow the model to be a critical tool in understanding unique urban dynamics.
- Accurate representations of urban features in numerical weather models are crucial for accurately simulating complex interactions between cities and the atmosphere.
- Accuracy of the bulk approach used in CONUS404 to simulate atmospheric conditions in urban areas at continental- and regional-scales remains unclear.
- Past studies have lacked a direct comparison of model errors & biases in urban grids to non-urban grids (Thompson et al., under review).
- These results have implications for improving the accuracy of simulated urban hydrometeorological conditions at different scales.

ii) Methodology

- Utilized the gap-filled Comprehensive Hourly Urban Weather Database (CHUWD-H v1.0) to derive hourly station-level, observed data
- Retrieved 10-years of CONUS404 2-m temperature, 2-m dewpoint, 10-m wind components (u & v), at 550 sites from the National Center for Atmospheric Research Research Data Archive
- Calculated mean absolute error (MAE)(tests accuracy) and mean bias error (MBE)(systematic bias) as CONUS404 – CHUWDH
- Applied and mapped seasonal analyses to gauge temporal performance
- Quantified the regionality of the model’s errors and biases using the nine NOAA Climate Regions (Karl and Koss, 1984)
- Tested the dependency of model error on the urban fraction of the model grid
- Verified results from (CONUS404-CHUWDH) with an independent dataset



vii) Key Points

- Temperature, dewpoint, & wind speed errors & biases are regionally constrained.
- Temperature in urban grids showed statistically significant poorer performance during the summer & a statistically significant warm MBE.
- Winter (Northeast) was the worst performing season, for all simulated variables assessed.
- Dewpoint dry biases are persistent in urban grids across all spatial & temporal scales.
- Magnitude of CONUS404 wind speed under-estimation is stronger in urban grids & is systematic across all spatial & temporal scales.

viii) References

Karl, T. R., and Koss, W. J. (1984). Regional and national monthly, seasonal, and annual temperature weighted by area, 1895–1983. National Climatic Data Center Historical Climatology Series, 4(3), 38 pp.

Rasmussen, R., Chen, F., Liu, C.H., Ikeda, K., Prein, A., Kim, J., Schneider, T., Dai, A., Gochis, D., Dugger, A., Zhang, Y., Jaye, A., Dudhia, J., He, C., Harrold, M., Xue, L., Chen, S., Newman, A., Dougherty, E., Abolafia-Rosenzweig, R., Lybarger, N.D., Viger, R., Lesmes, D., Skalak, K., Brakebill, J., Cline, D., Dunne, K., Rasmussen, K., Miguez-Macho, G., (2023). CONUS404: The NCAR-USGS 4-km long-term regional hydroclimate reanalysis over the CONUS. Bull. Am. Meteorol. Soc. 104, E1382–E1408. <https://doi.org/10.1175/BAMS-D-21-0326.1>

Thompson, L., Wang, C., He, C., Tzu-Shun, L., Changhai, L., & Dudhia, J., (under review). Assessment of convection-permitting hydroclimate modeling in urban areas across the conterminous United States. Manuscript under review in Urban Climate

Wang, C., Deng, C., Horsey, H., Reyna, J. L., Liu, D., Feron, S., Cordero, R. R., Song, J., & Jackson, R. B. (2024). CHUWD-H v1.0: a comprehensive historical hourly weather database for U.S. urban energy system modeling. Scientific Data, in press.

ix) Acknowledgements

We recognize the contributions provided by **Drs. Cenlin He, Tzu-Shun Lin, Changhai Liu, and Jimmy Dudhia**. Special recognition to Chenghao Wang for his active mentorship throughout the project. This work was supported by the **U.S. NSF** under grant Nos. **OIA-327435** and **CNS-2301858**, the **USGS** under grant No. **G24AC00475**, **NASA** under grant Nos. **80NSSC24K1056** and **80NSSC24K0357**, and **NOAA** under grant No. **NA21OAR4590361**. We also acknowledge support from the SoM, DGES, University of Oklahoma Honors College, the Undergraduate Research Opportunities Program (UROP), and the NCAR’s high-performance computing systems